

BiteFixes Enterprise Architecture Manual

Volume 13

Functional Module Specification (FMS)

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Table of Contents

1. Introduction
2. Functional Philosophy
3. Module Classification
4. Core Modules
5. Business Modules
6. AI Modules
7. Communication Modules
8. Financial Modules
9. Analytics Modules

- 10. Administration Modules
- 11. Integration Modules
- 12. Future Modules
- 13. Functional Dependencies
- 14. Evolution Strategy

Chapter 1

Introduction

The Functional Module Specification defines every functional capability of the BiteFixes platform.

Each module is independently deployable, independently testable, and independently evolvable.

Chapter 2

Functional Philosophy

Every module must answer four questions:

- What business problem does it solve?
- Which domains does it own?
- Which APIs does it expose?
- Which events does it publish and consume?

Chapter 3

Functional Classification

The platform is divided into six major groups.

Platform

- └─ Core Modules
- └─ Business Modules
- └─ AI Modules
- └─ Financial Modules
- └─ Communication Modules
- └─ Administration Modules

Chapter 4

Core Modules

These modules are mandatory.

Company Management

Responsibilities

- Company Registration
- Company Configuration
- Branding
- Subscription
- Localization

Identity & Access Management

Responsibilities

- Authentication
- Authorization
- MFA
- Sessions
- API Tokens

Customer Management

Responsibilities

- Customer Registration
- Customer Profile
- Preferences
- Customer History
- Customer Segmentation

Device Management

Responsibilities

- Device Inventory
- Device History
- IMEI
- Serial Numbers
- Warranty

Ticket Management

Responsibilities

- Ticket Lifecycle

- Assignment
- Workflow
- SLA
- Status Tracking

Chapter 5

Business Modules

Service Catalog

Responsibilities

- Categories
- Services
- Pricing
- Warranty Policies
- Estimated Times

Inventory

Responsibilities

- Parts
- Stock
- Warehouses
- Reservations
- Suppliers

Estimates

Responsibilities

- Cost Calculation
- Customer Approval
- Version History
- PDF Generation

Repair Management

Responsibilities

- Diagnosis
- Repair Process
- Quality Control
- Delivery

Warranty

Responsibilities

- Warranty Registration
- Claims
- Validation
- Expiration

Chapter 6

AI Modules

Bitey Core

Responsibilities

- Intent Detection
- Context Management

- Reasoning
- AI Orchestration

Knowledge Management

Responsibilities

- Knowledge Articles
- Approval
- Search
- RAG Preparation

AI Memory

Responsibilities

- Customer Memory
- Company Memory
- Conversation Memory
- Global Memory

AI Skills

Examples

- Printer Support
- Computer Repair
- Mobile Repair
- Networking
- Cloud
- Cybersecurity

Each skill is an independent package.

Chapter 7

Communication Modules

Chat

Channels

- Website
- Mobile
- WhatsApp

Notification Center

Channels

- Email
- WhatsApp
- Push
- SMS

Conversation History

Stores

- Messages
- Attachments
- AI Responses
- Technician Notes

Chapter 8

Financial Modules

Billing

Responsibilities

- Invoices
- Taxes
- Discounts
- Payments

Cash Register

Responsibilities

- Daily Cash
- Expenses
- Income
- Reconciliation

Accounting Integration

Responsibilities

- Export
- ERP Integration
- Tax Reports

Chapter 9

Analytics Modules

Dashboard

KPIs

- Revenue
- Tickets
- Repairs
- AI Usage
- Customer Satisfaction

Operational Reports

Examples

- Open Tickets
- Average Repair Time
- Technician Productivity
- Device Statistics

AI Analytics

Metrics

- Confidence
- Latency
- Knowledge Usage
- Escalation Rate

Chapter 10

Administration Modules

Workflow Designer

Visual workflow editor.

Policy Manager

Policy configuration.

Business Rule Manager

Business rule administration.

User Administration

Roles

Permissions

Groups

Teams

Audit Center

Searchable audit records.

Chapter 11

Integration Modules

REST API

Webhooks

Import/Export

CSV

Excel

JSON

External Services

Examples

- Payment Gateways
- Shipping Providers
- ERP Systems
- CRM Systems

Chapter 12

Future Modules

The architecture anticipates future capabilities.

Examples:

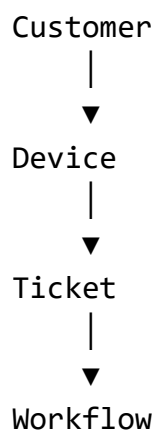
- Predictive Maintenance.
- IoT Monitoring.
- Remote Device Diagnostics.
- Fleet Management.
- Customer Self-Service Portal.
- AI Voice Assistant.
- Business Intelligence Studio.
- Marketplace for Third-Party Extensions.

These modules should integrate through the existing APIs and Event Bus without changing the core platform.

Chapter 13

Functional Dependencies

Dependencies should remain minimal.





Billing

Cross-module communication should occur through service contracts or events rather than direct database access.

Chapter 14

Evolution Strategy

Every module evolves independently.

Recommended lifecycle:

Prototype



Beta



Stable



Enterprise



Legacy



Retired

This allows new functionality to mature without disrupting existing installations.

Functional Map of BiteFixes

BiteFixes Platform

- Company Management
- Identity & Access
- Customer Management
- Device Management
- Ticket Management
- Service Catalog
- Inventory
- Estimates
- Repair Management
- Warranty
- Bitey Core
- Knowledge Management
- AI Memory
- AI Skills
- Chat
- Notifications
- Conversation History
- Billing
- Cash Register
- Accounting Integration
- Dashboards
- Reports
- AI Analytics
- Workflow Designer
- Policy Manager
- Business Rule Manager
- Audit Center
- REST API
- Webhooks
- Import/Export
- External Integrations